

Introduction to Viewing Astronomy Data: DS9

You should work together with a partner on all in class activities and have discussions, but each person should work on their own computer, and complete their own worksheet.

*Tasks to perform which don't require a response are in **green font**, while specific items requiring a response are in **bold red font**. I suggest making your own PD document where you record those answers, which you will upload when completed.*

Credit: This activity is adapted from “A Beginner’s Guide to Working with Astronomy Data” (BG), S3.0-3.3

Learning Objectives: Explore the data behind astronomy images. Be able to find and download data from the Hubble legacy archive. Learn how to open images, scale and zoom, and check sky coordinates using DS9.

The application software we will use in this section is called SAOImage DS9. DS9 is one of the most commonly-used FITS image viewing programs. It is public software developed at the Smithsonian Astrophysical Observatory (SAO) [see HW1]. It is already installed on our astronomy computers.

Getting started on your classroom computer.

1. **Log in!** Use your EID and password to log in to the computer.

Note: any files you store here are stored on this computer's hard drive so you won't be able to access them from another computer. For the semester, this will be your computer. The first time you log in you may need to set up your mouse options to right click and zoom (see System Preferences).

2. **Make a folder.** IT requests that all students organize their work for the semester in one location. Create a folder on the desktop titled **astrolab** (all lowercase, and no space). You will save all your work to this folder throughout the semester. For organization, I recommend making sub-folders, e.g., ClassActivities, HW, etc. *Even if you use your own computer, I strongly suggest forming an organized set of folders!*

Retrieving the data.

3. **Data archive:** We will get our astronomy image file from the Hubble legacy archive at the Space Telescope Science Institute (STScI) in Baltimore, which operates the Hubble Space

Telescopes and other space telescopes. The legacy archive is where all the older Hubble images are stored.

Go to the URL: <https://hla.stsci.edu/hlaview.html>. Once you load the page you will see there is a helpful **Search** box.

Search for M16, the Eagle Nebula, by entering “M16” in the search field and pressing **Search**.

4. **Inspect the available data:** You will see a long list of data — all observations of the Eagle Nebula taken by HST on different dates. Look in column 14 under **Dataset** and find: hst_05773_05_wfpc2_f502n_wf.

* **What date was this data taken (StartTime)?**

* **How long was the observation (ExpTime)?**

* **At what sky coordinates was HST pointed (RA, DEC)?**

If you know astronomical abbreviations, you can make some sense of these dataset names: hst, means that the data is from the Hubble Space Telescope. WFPC2 is the “Wide-Field and Planetary Camera 2” on that telescope, wf says that we are downloading the wide-field camera images. f502, f656 and f673 denote different filters which have been placed before the camera for these respective images.

5. **Get the data:** Once you find the right row of data, click the little blue FITS shopping cart for the image to add this data to your cart. Then click on the shopping cart tab at the top of the page. You will see this tab tells you the number of data sets you’ve selected and the total size of the data you’ve selected for download — important information so you don’t “buy” more data than your computer can handle!

Then click **Fetch HLA Data**. The data will be automatically downloaded to the Downloads folder on the computer. Find it and drag it to your folder.

Visualizing the data in **DS9**

6. **Launch DS9:** Open DS9 (e.g., type ds9 into the spotlight search box in the top right). DS9 has both a text menu bar and a button bar. The button bar has commonly-used options and functions. The text menu bar contains all options.

7. **Open a File:** To open the file do: File -> Open -> Filename

File will be in the main menu (first) row (it is probably highlighted to begin with). From the secondary menu that appears directly below, choose Open. In the pop-up choose-a-file window, choose the HST image we downloaded, hst 05773 05 wfpc2 f502n wf drz.fits. (Note your Mac may have automatically unzipped the downloaded file, in which case you might find the fits file inside a directory of the same name; you may also be able to double click on the image file and the Mac will open it in ds9 automatically).

8. **Visualize the Data:** The image you see should be disappointingly black. Don't panic! We need to find a better brightness scale to see what is going on. Astronomical images typically capture an amazing dynamic range, that is, an amazing range of different brightness values for each pixel (concretely, 65536 different brightness values per pixel, compared with the 256 of a typical Red-Green-Blue (RGB) pixel). Displaying such an image on a computer monitor, or printing it, can never do this range full justice. Instead, we need to pick and choose — which part of the brightness range do we want to display, and which way of compressing the brightness scale shows us the most information about the image?

Experiment: Go to the buttonbar and choose scale. From the second line of buttons, try different options. Which gives the “best” results?

Some parts of the image are darker and some brighter — what does this suggest about the emitting gas?

Note: There is no single right way of scaling the image, and there is no standard way that will guarantee the best results for all possible astronomical pictures. Instead, this is a matter of combining experience with some experimentation to arrive at a result that works for you. You should always be aware that such a result is not a naked view of the astronomical data — what you see is determined both by the astronomical data and by the choices you have made in

displaying that data. Also note that these display options do not change the image itself; the image file itself is unchanged, and you are only changing the way you are viewing the data.

9. **Scale the brightness:** From the secondary menu, choose **linear** and **zscale**. The default option (min-max) scales the color bar to display the image with the minimum pixel brightness set to black, and the maximum pixel brightness to white. **Zscale** scales the colors so the values are closer to the median pixel brightness. This still looks fairly raw, and different from the pretty astronomy pictures you see in the media. But at least you should be able to see some structures. **Experiment with different options under the color menu.** These map the data to a color table. Here, all options create “false color” image since these are not the true colors of the light.

10. **Look more closely:** Click on the “zoom” button, and zoom in a few times. In the overview window (second to right, on top) you can see the entire image. The cyan frame in the overview window marks the part of the image that is visible in the main window; after you zoom, it should be smaller than the full image. **Drag the overview window around (left-click the mouse and drag) to explore other parts of the image.** Center your image on the top of the middle pillar.

11. **Inspect coordinates:** Place your mouse cursor over the top of this pillar. You should see the coordinates in the list of properties change. The numbers give both the sky coordinates (RA, DEC) of the location of your cursor but also the pixel value (X, Y) within the image. The top right shows a zoomed in image of the data where your cursor is.

Find the object located at (RA, DEC = 18:18:51.78, -13:50:22.) This is a young star with a protoplanetary disk. **Place a region around this source.** You can do this by clicking the edit menu at the top of your screen and selecting region, then click on the source (now each time you click a region will appear).

Go to WCS in the top menu and change the notation from sexagesimal format (hours:minutes:seconds) to degrees. The latter will show both right ascension and declination as a decimal number denoting degrees. **Write here the RA and Dec in decimal degrees for this source?**

Given its coordinate (and light pollution aside), can we observe this young star from Austin?

12. **Look up the Eagle Nebula on wikipedia. What are three things you learned about this object?**

12. Bonus (if done early):

Go to ceers.github.io, click on Data Releases, then release v0.5. Download the F200W image for pointing NIRC2. Go to NASA ADS, find my paper on Maisie's Galaxy (titled: "A Long Time Ago..."). Find the coordinates for this object, find it in the image, and put a region around it.

Still have more time? Download the F115W image for NIRC2 and find this same location. Can you see the galaxy?